

MongoDB Sharding & Replica Set MCQ Questions

Section A: Sharding Fundamentals

Question 1: What is the primary purpose of MongoDB sharding?

- A) To create backup copies of data
- B) To horizontally scale data across multiple servers
- C) To compress data for storage efficiency
- D) To encrypt data for security

Answer: B

Explanation: Sharding distributes data horizontally across multiple servers (shards) to handle large datasets and high throughput operations.

Question 2: In a MongoDB sharded cluster, what is the role of mongos?

- A) Stores actual data documents
- B) Maintains cluster metadata
- C) Acts as a query router between clients and shards
- D) Performs data replication

Answer: C

Explanation: Mongos is the query router that interfaces between client applications and the sharded cluster, directing queries to appropriate shards.

Question 3: Which component stores metadata about chunk distribution in a sharded cluster?

- A) Mongos router
- B) Primary shard
- C) Config servers
- D) Secondary shards

Answer: C

Explanation: Config servers store metadata about the cluster including chunk ranges, shard locations, and cluster configuration.

Section B: Shard Key Strategies

Question 4: What happens when you use a hashed shard key?

- A) Data is distributed based on ranges of the shard key values
- B) Data is distributed based on hash values of the shard key
- C) Data is stored in alphabetical order
- D) Data is replicated to all shards

Answer: B

Explanation: Hashed sharding uses a hash function on the shard key value to distribute data evenly across shards, preventing hotspots.

Question 5: Which shard key strategy provides the best write distribution but doesn't support range queries efficiently?

- A) Range-based sharding
- B) Zone-based sharding
- C) Hashed sharding
- D) Compound sharding

Answer: C

Explanation: Hashed sharding distributes writes evenly but cannot efficiently support range queries since hash values don't preserve order.

Question 6: In range-based sharding, documents with shard key values from 1-100 go to shard1, 101-200 to shard2. What is this called?

- A) Hash distribution
- B) Round-robin distribution
- C) Zone/Tag-aware sharding
- D) Random distribution

Answer: C

Explanation: This describes zone-based or tag-aware sharding where specific ranges are assigned to specific shards using tags.

Section C: Replica Sets

Question 7: What is the minimum number of nodes required for a MongoDB replica set to maintain high availability?

- A) 1
- B) 2
- C) 3
- D) 5

Answer: C

Explanation: A minimum of 3 nodes is required to maintain majority for automatic failover (2 data-bearing nodes + 1 arbiter, or 3 data-bearing nodes).

Question 8: In a replica set, what happens during primary node failure?

- A) The cluster becomes read-only
- B) All operations stop until manual intervention
- C) A secondary node is automatically elected as primary
- D) Data is lost permanently

Answer: C

Explanation: MongoDB replica sets automatically elect a new primary from available secondary nodes when the current primary fails.

Question 9: Which read preference allows reading from secondary nodes?

- A) primary
- B) primaryPreferred
- C) secondary
- D) All of the above

Answer: D

Explanation: 'primary' reads only from primary, 'primaryPreferred' prefers primary but can read from secondary, 'secondary' reads only from secondary nodes.

Section D: Configuration and Setup

Question 10: What command enables sharding on a database?

- A) `sh.enableSharding("dbName")`
- B) `db.enableSharding()`
- C) `mongos.enableSharding("dbName")`
- D) `rs.enableSharding("dbName")`

Answer: A

Explanation: The `sh.enableSharding("databaseName")` command enables sharding on a specific database.

Question 11: To create a sharded collection with hashed shard key on `userId`, which command is correct?

- A) `sh.shardCollection("db.users", { userId: 1 })`
- B) `sh.shardCollection("db.users", { userId: "hashed" })`
- C) `sh.createShard("db.users", { userId: "hash" })`
- D) `db.users.shard({ userId: "hashed" })`

Answer: B

Explanation: `sh.shardCollection("db.collection", { field: "hashed" })` creates a hashed shard key on the specified field.

Question 12: In the setup script, what does `rs.initiate()` do?

- A) Starts the MongoDB service
- B) Initializes a replica set configuration
- C) Creates a new database
- D) Enables sharding on a collection

Answer: B

Explanation: `rs.initiate()` initializes a replica set with the specified configuration, making nodes aware of each other.

Section E: Zone-Based Sharding

Question 13: What is the correct sequence to implement zone-based sharding?

- A) Create zones → Add shard tags → Define ranges → Shard collection
- B) Shard collection → Add shard tags → Create zones → Define ranges
- C) Add shard tags → Define ranges → Shard collection
- D) Define ranges → Shard collection → Add shard tags

Answer: C

Explanation: First add tags to shards, then define tag ranges, and finally shard the collection. Zones are created implicitly when adding tags.

Question 14: Which command assigns a zone tag to a shard?

- A) `sh.addZone("shardName", "zoneName")`
- B) `sh.addShardTag("shardName", "tagName")`
- C) `sh.createZone("shardName", "zoneName")`
- D) `sh.tagShard("shardName", "tagName")`

Answer: B

Explanation: `sh.addShardTag("shardName", "tagName")` assigns a zone tag to a specific shard.

Question 15: To route documents with `userId` 1-100 to zone1, which command is used?

- A) `sh.addTagRange("db.users", {userId: 1}, {userId: 100}, "zone1")`
- B) `sh.addTagRange("db.users", {userId: 1}, {userId: 101}, "zone1")`
- C) `sh.createRange("db.users", 1, 100, "zone1")`
- D) `sh.setRange("db.users", {userId: 1-100}, "zone1")`

Answer: B

Explanation: Range boundaries are exclusive on the upper bound, so `{userId: 101}` ensures documents with `userId` 1-100 go to zone1.

Section F: Performance and Scaling

Question 16: What is a chunk in MongoDB sharding?

- A) A backup copy of data
- B) A contiguous range of shard key values
- C) A single document
- D) A database index

Answer: B

Explanation: A chunk is a contiguous range of shard key values that MongoDB uses to distribute data across shards.

Question 17: When does MongoDB automatically split chunks?

- A) When chunk size exceeds the configured limit (default 64MB)
- B) Every hour automatically
- C) When manually triggered by administrator
- D) When a new shard is added

Answer: A

Explanation: MongoDB automatically splits chunks when they exceed the configured chunk size limit (default 64MB) to maintain balanced distribution.

Question 18: What is the balancer in MongoDB sharding?

- A) A tool to balance CPU usage
- B) A process that migrates chunks between shards to maintain even distribution
- C) A load balancer for client connections
- D) A backup scheduling system

Answer: B

Explanation: The balancer is a background process that automatically migrates chunks between

shards to maintain even data distribution.

Section G: Multiple Mongos and Load Distribution

Question 19: Can you have multiple mongos routers in a sharded cluster?

- A) No, only one mongos is allowed per cluster
- B) Yes, multiple mongos can connect to the same cluster for load distribution
- C) Yes, but each mongos must connect to different shards
- D) Only in replica set configurations

Answer: B

Explanation: Multiple mongos routers can connect to the same sharded cluster, providing load distribution and high availability for client connections.

Question 20: What is the advantage of running multiple mongos instances?

- A) Increases storage capacity
- B) Provides connection load balancing and eliminates single point of failure
- C) Improves data compression
- D) Enables automatic backups

Answer: B

Explanation: Multiple mongos instances distribute client connection load and eliminate the mongos router as a single point of failure.

Section H: Advanced Configuration

Question 21: In a production sharded cluster, how many config servers are recommended?

- A) 1
- B) 2
- C) 3
- D) 5

Answer: C

Explanation: 3 config servers are recommended for production to provide high availability and fault tolerance for cluster metadata.

Question 22: What happens if all config servers in a sharded cluster fail?

- A) Data is lost permanently
- B) Cluster becomes read-only but existing connections continue
- C) New connections cannot be established and metadata operations fail
- D) Automatic failover to backup config servers

Answer: C

Explanation: Without config servers, new connections cannot be established and metadata operations (like chunk migrations) fail, though existing connections may continue.

Question 23: Which MongoDB version introduced automatic chunk splitting?

- A) MongoDB 2.6
- B) MongoDB 3.4
- C) MongoDB 4.0
- D) MongoDB has always had automatic chunk splitting

Answer: D

Explanation: MongoDB has included automatic chunk splitting since early versions as a core feature of the sharding system.

Section I: Troubleshooting and Monitoring

Question 24: Which command shows the current sharding status and chunk distribution?

- A) `sh.status()`
- B) `db.shardStatus()`
- C) `rs.status()`
- D) `db.stats()`

Answer: A

Explanation: `sh.status()` displays comprehensive information about sharding configuration, shards, and chunk distribution.

Question 25: How can you check if a collection is sharded?

- A) `db.collection.isSharded()`
- B) `sh.status()` and look for the collection
- C) `db.collection.getShardDistribution()`
- D) Both B and C

Answer: D

Explanation: Both `sh.status()` shows all sharded collections and `db.collection.getShardDistribution()` shows distribution for a specific collection.

Answer Key Summary:

1. B 2. C 3. C 4. B 5. C
2. C 7. C 8. C 9. D 10. A

3. B 12. B 13. C 14. B 15. B

4. B 17. A 18. B 19. B 20. B

5. C 22. C 23. D 24. A 25. D

Scoring Guide:

- 23-25 correct: Expert level
- 20-22 correct: Advanced level
- 17-19 correct: Intermediate level
- 14-16 correct: Beginner level
- Below 14: Review fundamentals needed